

# A Lightweight LLM-Driven Multi-Agent Framework for Virtual Society Simulation with Memory Feedback

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*Valentown — seven residents who plan, meet, remember and adapt*

# Two ways to simulate people

## **R** Rule-based agent models

Fixed rules



Action

*same input → the same behaviour, every day*

No context · no planning · no memory

Macro patterns yes — human-like behaviour no

## **L** LLM-driven agents

Prompt



LLM



Action

*fluent — but one step at a time*

Reactive: answers the prompt in front of it

Interaction ad hoc or random · no day structure

**Missing: explicit structure for behaviour, interaction and feedback**

# One question, three requirements

*How can we design a virtual society whose agents are structured, evolving and controllable over time?*

## 1 Temporal structure

A day — not a stream of reactions

### OUR ANSWER

**Schedule-driven agents**

## 2 Behavioural evolution

Yesterday changes tomorrow

### OUR ANSWER

**Memory feedback loop**

## 3 Controllability

An explicit rule, plus a human override

### OUR ANSWER

**Co-location trigger + human-in-the-loop**

# A backend that thinks, a frontend that shows

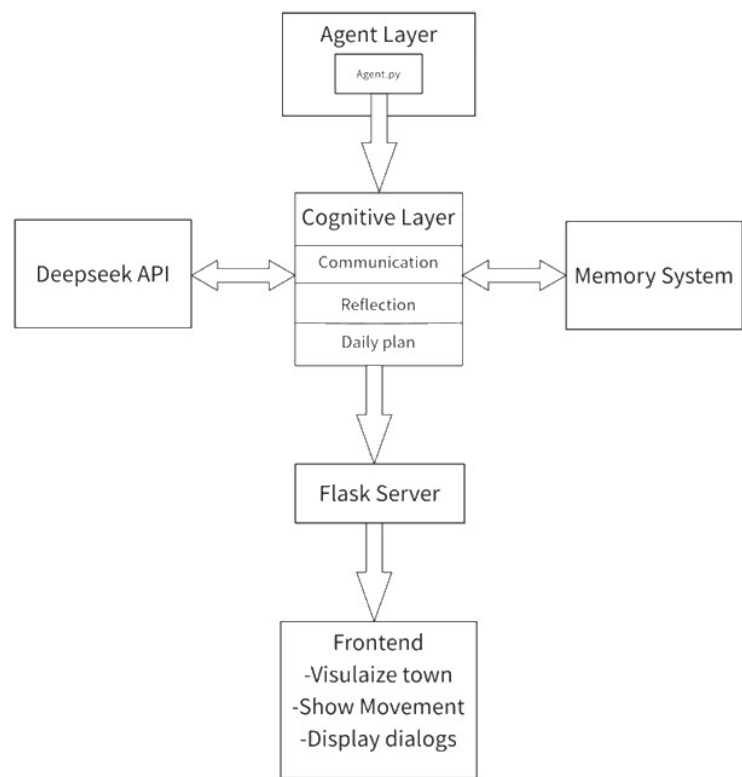


Fig. 1 Overview of the proposed framework

## Backend · Python

Planning, dialogue, reflection, memory — the only part that calls the LLM

## Frontend · Phaser

Town, movement and conversations, served over a REST API

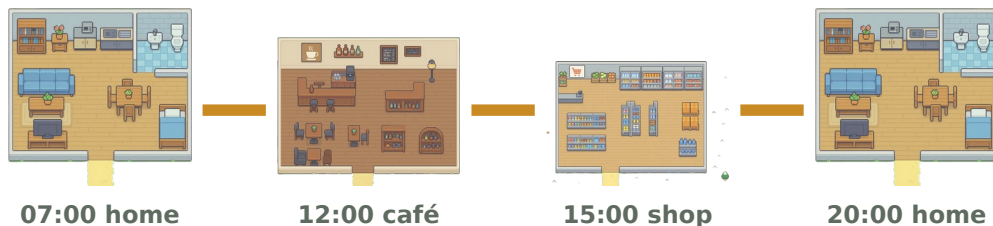
## Five modules

Agent Manager · Scheduler · Interaction Engine · LLM Interface · Memory Manager

# Plan the day, then let the town decide who meets

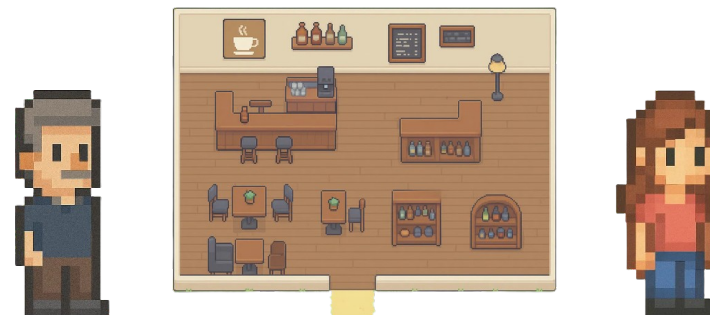
## 1 Schedule-driven day

identity + goals + memory → one day's plan



*one generated day, illustrative*

## 2 Co-location interaction



same place + same period → they talk

**Deterministic when · generative what**

the framework controls the dynamics — the model supplies the diversity

# Memory is where the days connect

## 3 Memory-based feedback loop

### WRITTEN IN

Plans

Conversations

Reflections

### READ OUT

Planning

Dialogue

Reflection

Every entry: category · timestamp · importance score  
Retrieval: recent + important · the store stays bounded

### OPTIONAL: HUMAN IN THE LOOP

Feedback written into the same memory channel

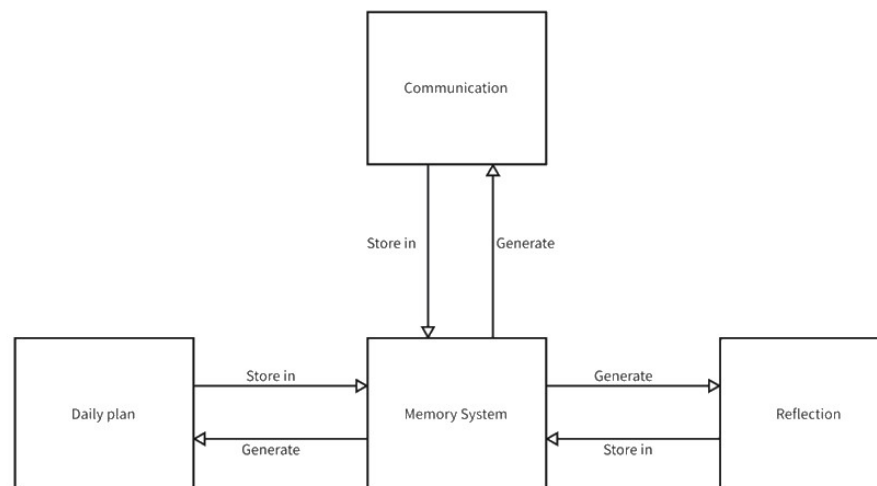


Fig. 3 Memory-centred feedback loop

**The LLM stays stateless — all continuity lives in memory**

# One simulation day, end to end

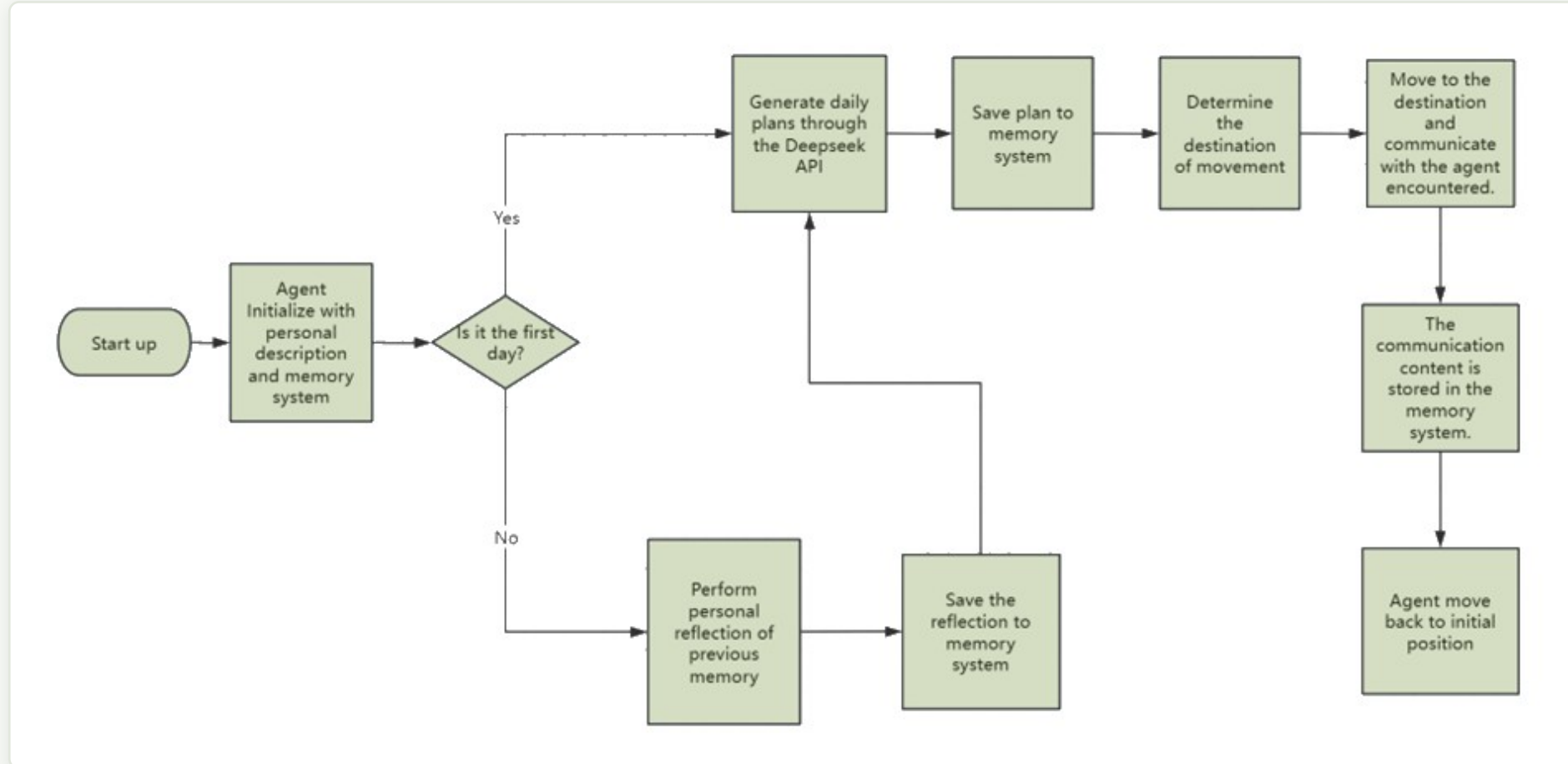


Fig. 2 · Day 1 plans from the profile — every later day reflects first, then plans

# Valentown: the framework, running



## Valentown

homes · shop · pharmacy · café · park

## 7 residents

own age, role, personality, goals

## DeepSeek API

schedules · dialogue · reflection

## Python + Phaser

pre-computed, replayed over REST



## EXPERIMENTS

# What we ran, and what we looked for



**Ron Parker**

60 · supermarket owner



**Ella Parker**

58 · pharmacy owner



**Emma Harris**

30 · mother



**Gavin Harris**

32 · father



**Adam Harris**

7 · child



**Mia Thompson**

28 · family teacher



**Arthur Morgan**

29 · architect

### Environment

predefined locations;  
co-location enables interaction

### Duration

multiple consecutive days,  
a full cycle each day

### Execution

deterministic triggers,  
stochastic content

## WE LOOK AT

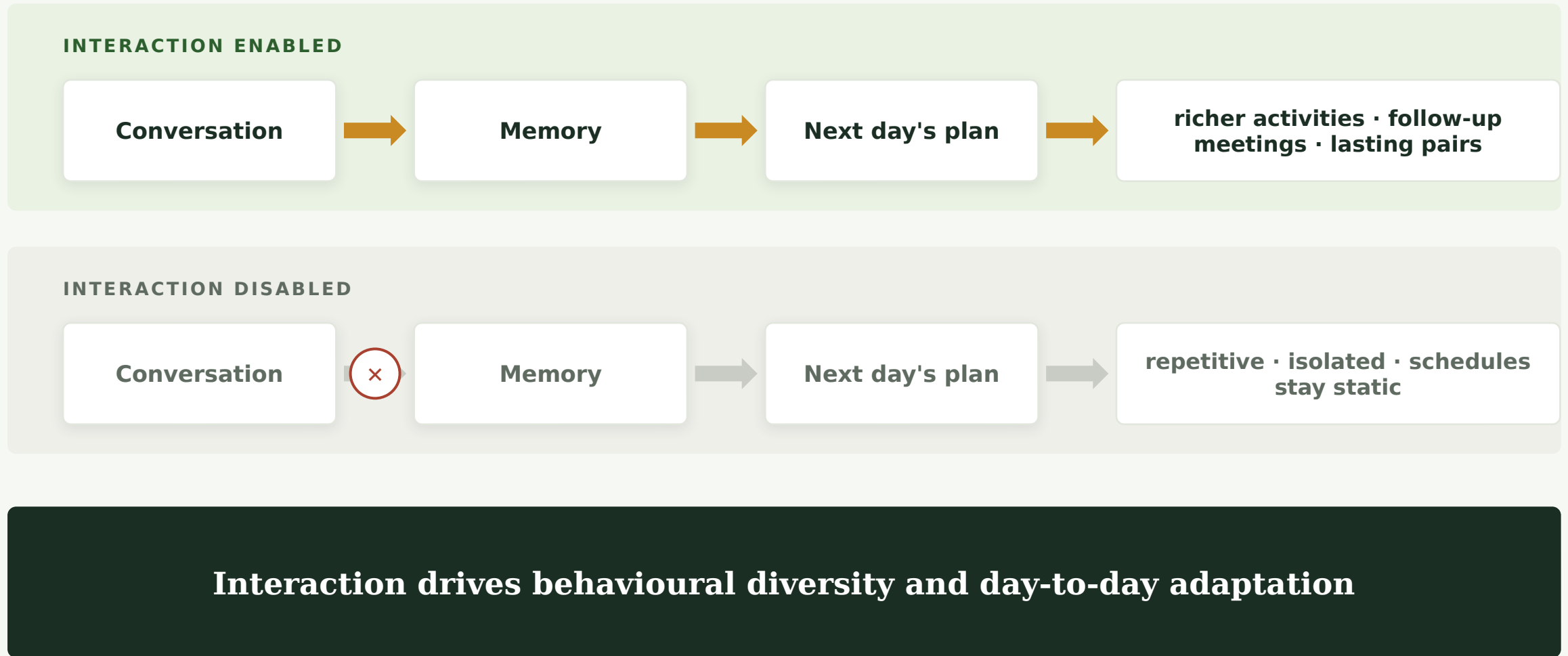
**interaction → behaviour**

**memory → adaptation**

**emergent social structure**

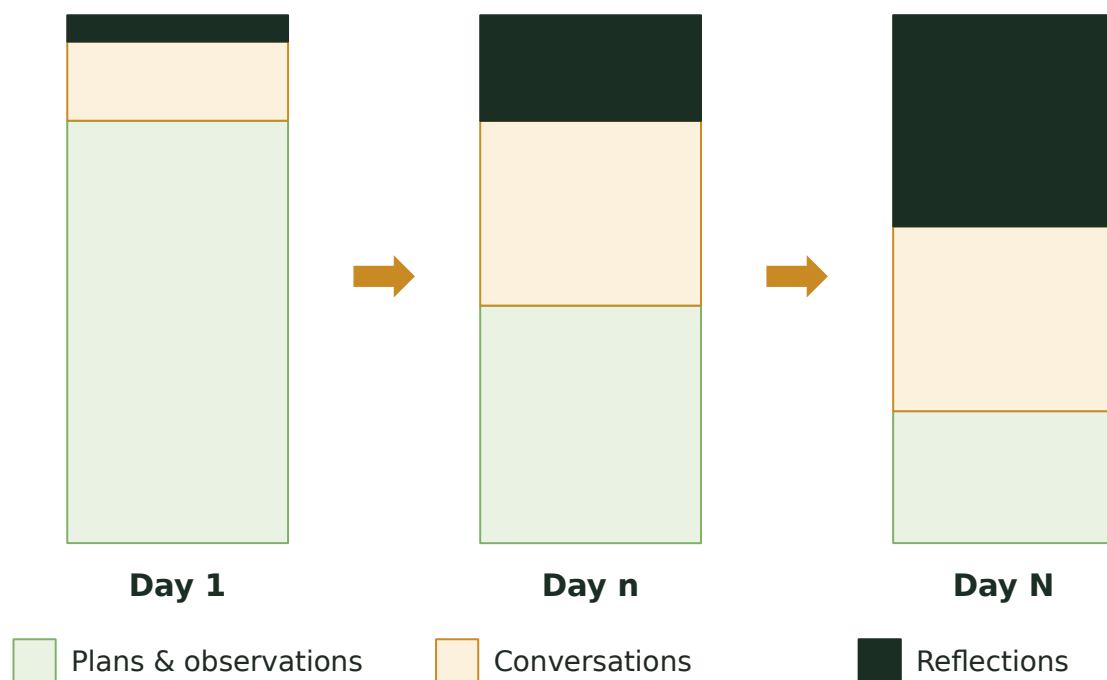
## RESULT 1

# Turn interaction off, and the town goes quiet



# Memory changes shape as the days pass

## What memory is made of



*schematic — relative composition, not measured proportions*

## Bonds form on their own

Repeated meetings → persistent pairs → repeated co-location.

Nobody wrote a friendship rule.

## Behaviour follows memory

Reflection-derived insight → goal-aligned task selection shifts across days.

Limit memory, and adaptation weakens.

# What this buys us, and what it costs

## Advantages

- 1 Lightweight & reproducible**  
explicit pipeline, deterministic triggers
- 2 Controllable**  
schedules, co-location rules, human feedback
- 3 Extendable**  
every module swaps independently



*any module can be swapped without touching the others*

## Limitations

- 1 Cost**  
API calls scale with agents × days
- 2 Scale**  
small population, simple town
- 3 Stochasticity**  
runs vary — strict metrics are hard



*every plan, conversation and reflection is an API call*

# Where this leaves us

## What we showed

- 1 Planning, interaction and reflection linked by one memory loop
- 2 Importance-based memory + reflection → next-day behaviour
- 3 Consistent routines, lasting bonds, adaptation — no hand-written rules



## Future work

- 1 **Quantitative evaluation**  
network metrics, plan coherence, user studies
- 2 **Scale & real time**  
larger populations, live reaction
- 3 **Deeper reasoning**  
NPCs, education, embodied agents

# Thank you

Questions are very welcome.

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